
Algorithm 1 C-MVH-MIN*: refinement of Min-CTDC

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1: insert start node using key  $\mu(g + h_{\text{id}}(s))$ 
2: while OPEN is nonempty and Skyler's Done condition is false do
3:    $n = (s, g) \leftarrow \text{extract-Min}(\text{OPEN})$ 
4:   if Skyler-CTDC-prune( $g + h_{\text{id}}(s), T$ ) then
5:     continue
6:   if not SURVIVES( $s, g, T$ ) then
7:     continue ▷ new, strictly stronger pruning
8:   expand  $n$  and insert successors with ideal-point Min keys

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Compact Multi-Valued Heuristics under Min Ordering

Refined idea and preliminary evidence for bi-objective search

1. Question and refined contribution

Skyler et al. show that bi-objective search under the **Min** ordering admits constant-time dominance checks (CTDC). Their algorithm nevertheless uses one single-valued heuristic (SVH), normally the *ideal point*. Our question is whether a multi-valued heuristic (MVH) can add useful pruning without losing Min's CTDC advantage or creating one search node per heuristic vector.

C-MVH-MIN* keeps **exactly the same ordering key and CTDC mechanism as Skyler et al.**, but adds an indexed set-level test over $H(s)$. One physical path is represented by one node (s, g) , irrespective of $|H(s)|$. Therefore the contribution is not a replacement for Min-CTDC; it is a strict pruning extension of it:

Skyler: Min key + CTDC + ideal-point pruning $\longrightarrow$ C-MVH-MIN*: the same machinery + MVH survival pruning.

2. Why Min permits compact MVH ordering

For a state s , let $H(s) \subseteq \mathbb{R}_{\geq 0}^2$ be a set of non-dominated admissible heuristic vectors and define its ideal point

$$h_{\text{id}}(s) = \left(\min_{h \in H(s)} h_1, \min_{h \in H(s)} h_2 \right).$$

Let $\mu(f) = \min(\bar{f}_1, \bar{f}_2)$ be the primary Min value after coordinatewise increasing normalization. Then

$$\min_{h \in H(s)} \mu(g + h) = \mu(g + h_{\text{id}}(s)).$$

Thus a node representing the entire set $\{g + h : h \in H(s)\}$ receives its exact primary Min value in $O(1)$ from $h_{\text{id}}(s)$. No heuristic member must be selected to place the node in OPEN. This is the key difference from existing Lex-based MVH algorithms, which bind a node to a concrete surviving h .

3. Strictly stronger pruning than the original Min algorithm

Let T be the currently known solution frontier. Skyler's SVH test prunes when

$$\exists t \in T : \quad t \preceq g + h_{\text{id}}(s).$$

Because $h_{\text{id}}(s) \preceq h$ for every $h \in H(s)$, this implies that every $g + h$ is dominated. The converse is false: $g + h_{\text{id}}(s)$ can survive even when *every realizable MVH evaluation* is dominated. We therefore query

$$\text{SURVIVES}(s, g, T) = [\exists h \in H(s) : \nexists t \in T, t \preceq g + h].$$

If it returns false, the node cannot complete to a new Pareto-optimal solution. Consequently, relative to Skyler's algorithm with the same OPEN ordering, C-MVH-MIN* can never expand more nodes due to pruning, and sometimes expands strictly fewer; its only possible disadvantage is the cost and memory of the additional oracle.

4. Exact indexed survival query

In two dimensions, $H(s)$ is a sorted staircase. A balanced tree stores for each contiguous subtree N its componentwise lower corner

$$\text{lo}(N) = \left(\min_{h \in N} h_1, \min_{h \in N} h_2 \right).$$

If T dominates $g + \text{lo}(N)$, it dominates every member of N , so the entire subtree is discarded. At a leaf, an undominated stored member is a valid survival witness. The query is exact; its worst case remains linear, but it is output-sensitive and can terminate after very few corner tests.

5. Proof-of-concept methodology

We evaluated the *maximum plausible pruning benefit* before implementing a full production solver.

- Data: real DIMACS New York road graph, with distance and travel time as the two objectives; connected subgraphs of 30k, 60k, and 120k states.
- We computed $H^*(s)$, the exact cost-to-go Pareto frontier at every state. This is a perfect MVH and therefore gives an unattainable but informative upper bound.

- Finite admissible MVHs of sizes $k = 2, 4, \dots, 128$ were created by partitioning each $H^*(s)$ staircase into k contiguous groups and replacing each group by its componentwise minimum. Hence $k = 1$ is exactly Skyler’s ideal-point baseline, while $k = \text{full}$ is the perfect-MVH ceiling.
- For a clean ablation, all runs used the same graph, Min key, ideal point, CTDC-style SVH pruning, and solution handling. Only the MVH survival layer was enabled. All variants returned identical Pareto fronts.
- Eleven queries were deliberately selected across distance bands while favoring rich frontiers. They test the intended regime and are not claimed to be an unbiased random-query benchmark.

6. Preliminary results

Map	$ H^*(s_0) $ range	$k = 4$	$k = 8$	$k = 16$	Perfect MVH
30k	35–52	3.7–37.1%	3.8–47.4%	3.8–49.4%	3.8–49.8%
60k	53–282	8.3–41.4%	9.7–46.9%	10.7–48.9%	11.2–49.1%
120k	209–713	7.4–20.6%	8.7–23.5%	9.2–25.0%	9.7–25.6%
Mean over 11 queries		19.1%	22.2%	23.4%	23.8%

The table reports reduction in node expansions relative to the $k = 1$ ideal-point baseline. The main findings are:

- The MVH layer has material pruning value in the intended regime: the perfect-MVH ceiling averaged 23.8%, with individual reductions up to 49.8%.
- More heuristic vectors are more accurate, but the search benefit saturates sharply: mean savings were 12.2% at $k = 2$, 19.1% at $k = 4$, 22.2% at $k = 8$ and 23.4% at $k = 16$, against a 23.8% perfect-MVH ceiling. Thus $k = 8$ captured 93% and $k = 16$ captured 98% of the ceiling, and $k \geq 32$ added under 0.5%. The survival test returns a Boolean, so once a node’s whole set is certified dominated, refining the set further cannot remove any additional node.
- The corner tree used between 1.7 and 7.8 dominance checks per survival query (4.3 on average), even for a start frontier of 713 vectors. This supports the claim that the set test can be far cheaper than scanning $H(s)$.
- The last Pareto solution was found at 94.4–100% of expansions (mean 99.9%). Therefore the observed gain comes from *local MVH pruning*, not from global early termination: under Min the search is not dominated by a long unproductive tail that a stronger termination test could cut.
- **Memory is the binding constraint on scale.** A 220k-state run exceeded 7.6 GB while storing exact $H^*(s)$ for every state and had to be abandoned. This is a property of our oracle-based methodology rather than of C-MVH-MIN* itself, but it delimits what we can currently measure, and it signals that per-state MVH storage — not query time — is the first practical bottleneck a real implementation must address.

7. What is established, and what remains

These results show that the idea is **not redundant with Skyler et al.**: their ideal-point test leaves nodes that a full MVH can safely remove, and the gap can be large on rich bi-objective road queries. They also sharpen the paper’s message: thousands of vectors are not inherently necessary; the important property is obtaining most of the full-set pruning power with a compact, indexed representation.

This is still a viability result, not a complete end-to-end comparison. $H^*(s)$ is too expensive to use as a practical heuristic, and the grouped MVHs were derived from this oracle. A full evaluation must construct realistic MVHs (e.g., A*pex approximations), include preprocessing and memory, compare wall-clock time with Skyler’s implementation and Maya Wolff’s MVH algorithm, and sample queries without favoring rich frontiers. The decisive question is whether the saved expansions outweigh MVH construction, storage, and oracle cost. The current evidence is strong enough to justify that implementation.